

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

**Nama Produk** **Doxorubicin hydrochloride**  
**Product Description:** **Doxorubicin hydrochloride**  
**Cat No. :** BP2516-1; BP2816-10; BP2516-5; BP2516-50  
**Synonyms** cytotoxic drug; Adriamycin hydrochloride; Hydroxydaunorubicin hydrochloride  
**CAS-No** 25316-40-9  
**Molecular Formula** C<sub>27</sub>H<sub>29</sub>NO<sub>11</sub>.HCl

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

**Company** Fisher Scientific (M) Sdn Bhd No. 3, Jalan Sepadu 25/123,  
 Taman Perindustrian Axis, Seksyen 25,  
 40400 Shah Alam, Selangor Darul Ehsan, Malaysia.

**Supplier**  
**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number**  
 (603) 5122 8888

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                     |
|----------------------------------------------------|---------------------|
| Acute oral toxicity                                | Category 4 (H302)   |
| Skin Corrosion/Irritation                          | Category 2 (H315)   |
| Serious Eye Damage/Eye Irritation                  | Category 1 (H318)   |
| Germ Cell Mutagenicity                             | Category 1A (H340)  |
| Reproductive Toxicity                              | Category 2 (H361fd) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335)   |

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H302 - Harmful if swallowed  
 H315 - Causes skin irritation

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H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H340 - May cause genetic defects  
H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child

## Precautionary Statements

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician  
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Other Hazards

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                                                                                                                                                            | CAS-No     | Weight % |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| 5,12-Naphthacenedione,<br>10-[(3-amino-2,3,6-trideoxy-.alpha.-L-lyxo-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-8-(hydroxyacetyl)-1-methoxy-, hydrochloride, (8S-cis)- | 25316-40-9 | > 95     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

### Most important symptoms and effects, both acute and delayed

Causes eye burns. Causes severe eye damage.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep refrigerated.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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## Control Parameters

### Exposure Controls

#### **Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### **Personal protective equipment**

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

#### **Hygiene Measures**

When using do not eat, drink or smoke Provide regular cleaning of equipment, work area and clothing Avoid contact with skin, eyes or clothing Wash hands before breaks and immediately after handling the product Keep away from food, drink and animal feeding stuffs

#### **Environmental exposure controls**

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Appearance</b>     | Orange                   |
| <b>Physical State</b> | Powder Solid             |
| <b>Odor</b>           | Odorless                 |
| <b>Odor Threshold</b> | No data available        |
| <b>pH</b>             | No information available |

|                            |                          |
|----------------------------|--------------------------|
| <b>Melting Point/Range</b> | 216 °C / 420.8 °F        |
| <b>Softening Point</b>     | No data available        |
| <b>Boiling Point/Range</b> | No information available |
| <b>Flash Point</b>         | Not applicable           |

**Method -** No information available

|                                 |                          |       |
|---------------------------------|--------------------------|-------|
| <b>Evaporation Rate</b>         | Not applicable           | Solid |
| <b>Flammability (solid,gas)</b> | No information available |       |

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**Explosion Limits** No data available

|                                     |                          |       |
|-------------------------------------|--------------------------|-------|
| <b>Vapor Pressure</b>               | No data available        |       |
| <b>Vapor Density</b>                | Not applicable           | Solid |
| <b>Specific Gravity / Density</b>   | No data available        |       |
| <b>Bulk Density</b>                 | No data available        |       |
| <b>Water Solubility</b>             | Soluble                  |       |
| <b>Solubility in other solvents</b> | No information available |       |

**Partition Coefficient (n-octanol/water)**

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Autoignition Temperature</b>  | Not applicable           |       |
| <b>Decomposition Temperature</b> | No data available        |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| <b>Molecular Formula</b> | C <sub>27</sub> H <sub>29</sub> NO <sub>11</sub> .HCl |
| <b>Molecular Weight</b>  | 580.03                                                |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Light sensitive. Moisture sensitive.

### Possibility of Hazardous Reactions

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>      | None under normal processing.            |

### Conditions to Avoid

Incompatible products. Excess heat. Exposure to moisture. Exposure to light.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

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## Information on Toxicological Effects

### Acute Toxicity

### Chronic Toxicity

#### **Carcinogenicity**

The table below indicates whether each agency has listed any ingredient as a carcinogen

#### **Sensitization**

#### **Mutagenic Effects**

#### **Reproductive Effects**

#### **Developmental Effects**

#### **Teratogenicity**

#### **Target Organs**

No information available

Mutagenic effects have occurred in humans.

Experiments have shown reproductive toxicity effects on laboratory animals

Developmental effects have occurred in experimental animals

Teratogenic effects have occurred in experimental animals.

No information available.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Do not empty into drains.

### Persistence and degradability

No information available

### Bioaccumulative potential

No information available

### Mobility in soil

No information available.

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not dispose of waste into sewer

## SECTION 14: TRANSPORT INFORMATION

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|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>Road and Rail Transport</b>      | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component                                                                                                                                                                         | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|-----|------|-----|------|-------|------|-------|------|----------|
| 5,12-Naphthacenedione, 10-[(3-amino-2,3,6-trideoxy-.alpha.-L-lyxo-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-8-(hydroxyacetyl)-1-methoxy-, hydrochloride, (8S-cis)- | 246-818-3 | -      |     | -    | -   | -    | -     | -    | -     | -    | KE-01613 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

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BCF - Bioconcentration factor

VOC (volatile organic compound)

## Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**